

Research and adaptation of components from the persistent OS Phantom for Genode.

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Introduction

Data persistence: period of time during which data exists and is used

R. Morrison and M. P. Atkinson, "Persistent languages and architectures"

History of cryptography

Figure: Ceasar cipher

Figure: Skytale

Figure: Examples of ancient cryptographic machinery

First cryptographers

Figure: The first page of al-Kindi's manuscript "On Deciphering Cryptographic Messages", containing the oldest known description of cryptanalysis by frequency analysis.

Figure: Auguste Kerckhoffs

Figure: Claude Shannon

Modern cryptography

Figure: Whitfield Diffie and Martin Hellman, authors of the first published paper on public-key cryptography.

Figure: Ron Rivest, Adi Shamir and Leonard Adleman — reinventors of RSA cryptosystem.

Applications of cryptography

- ▶ Blockchains
- ▶ Secure messaging
- ▶ SSL\TLS
- ▶ Secure data storage
- ▶ You name it!

Why mathematical cryptography?

Selected topics in mathematics

- ▶ Abstract algebra (must have!)
- ▶ Number theory (must have!) — although we will only scratch the surface
- ▶ Probability theory
- ▶ Type theory
- ▶ Formal Logic

